



MISO DPP 2020 West Area Study Phase 1 Final Report

October 27, 2021

**MISO
720 City Center
Drive Carmel
Indiana 46032**
<http://www.misoenergy.org>

Contents

1. Executive Summary	4
1.1. Project List.....	4
1.2. Project Summary Network Upgrades.....	7
1.3. Total Network Upgrades	22
2. FERC Order 827 Compliance Review.....	29
3. Model Development and Study Assumptions.....	32
3.1. Base Case Models	32
3.2. Monitored Elements	32
3.3. Contingencies.....	32
3.4. Study Methodology	32
3.5. Performance Criteria	32
4. Backbone Network Upgrade Analysis	33
5. Thermal Analysis	33
6. Voltage Analysis	33
7. Stability Analysis	33
8. Short Circuit Analysis.....	33
9. Affected System Impact Study	33
10. Deliverability Analysis	33
10.1. Introduction.....	33
10.2. Determining the MW Restriction	34
10.3. Deliverability Study Results.....	34
11. Shared Network Upgrades Analysis	35
12. Cost Allocation	35
12.1. Cost Assumptions for Network Upgrades	35
12.2. Cost Allocation Methodology.....	35
Appendix A – Thermal Analysis Results	37
Thermal Constraints	37
Appendix B – Voltage Constraints	37
Voltage Constraints	37
Appendix C – Backbone Analysis Results	37
Appendix D – Network Upgrade Cost Allocation.....	37
Appendix E – MWEX Voltage Stability Report	37
Appendix F – Deliverability Tables.....	37
Appendix G – Deliverability Results	37
Table 1: List of DPP 2020 West Area Phase 1 Projects.....	4
Table 2: Total Cost of Network Upgrades for DPP 2020 West Phase 1 Projects	22
Table 3: ERIIS & NRIS Upgrades (Planning level cost estimates)	24



Table 4: Shared Network Upgrades (Planning level cost estimates)	28
Table 5: FERC Order 827 Review Results	29

1. Executive Summary

This report presents the results of a System Impact Study (SIS) performed to evaluate the interconnection of the generators in the DPP 2020 West Area Phase 1 (West Area DPP 1). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – American Transmission Company, Ameren, Basin Electric Power Cooperative, CIPCO, Cedar Falls, Dairyland Power Cooperative, Great River Energy, ITC Midwest, Montana-Dakota Utilities, MidAmerican, Minnesota Power, Ottertail Power, SMMPA, and Xcel Energy.

1.1. Project List

The original interconnection requests for DPP 2020 West Area had a total of 45 projects generation projects with a combined nameplate rating of 5432.5 MW (ERIS) & 5471 MW (NRIS). The detailed list of 2020 West Area DPP Phase 1 is shown below in Table 1, and the 2020 West cycle originally kicked off on January 6, 2021.

Table 1: List of DPP 2020 West Area Phase 1 Projects

Project ID	Fuel Type	TO	Service Request Type	Service Request (MW)	POI	Interconnection Self Fund Election
J1494	Battery Storage	Northern States Power (Xcel Energy)	NRIS	50	Chisago 115kV	Yes
J1495	Battery Storage	Northern States Power (Xcel Energy)	NRIS	50	North Rochester 161kV	Yes
J1496	Battery Storage	Southern Minnesota Municipal Power Agency	NRIS	50	Byron 161kV	No
J1498	Battery Storage	Northern States Power (Xcel Energy)	NRIS	50	Chanarambie 115 kV Substation	Yes
J1503	Solar	ITC Midwest	NRIS	150	Huntley 345 kV	Yes
J1504	Solar	ITC Midwest	NRIS	150	Huntley 345 kV	Yes
J1520	Solar	ITC Midwest	NRIS	135	Ledyard - Colby 345 kV	Yes
J1521	Wind	Otter Tail Power Company	NRIS	199.5	Big Stone - Ellendale 345kV	Yes
J1528	Battery Storage	Northern States Power (Xcel Energy)	ERIS	0*	King – Eau Claire 345 KV	Yes
J1529	Solar	MidAmerican Energy Company	NRIS	50	Carroll - Auburn 69 kV	Yes
J1530	Solar	MidAmerican Energy Company	NRIS	50	Hastings - Macedonia 69 kV	Yes
J1532	Solar	MidAmerican Energy Company	NRIS	50	Clarinda-Coburg 69 kV	Yes
J1533	Solar	MidAmerican Energy Company	NRIS	200	Creston-Adams County 161kV	Yes
J1534	Solar	Southern Minnesota Municipal Power	NRIS	100	Byron 345kV	No

Project ID	Fuel Type	TO	Service Request Type	Service Request (MW)	POI	Interconnection Self Fund Election
		Agency				
J1544	Solar	Montana-Dakota Utilities Company	NRIS	200	South Mandan 230kV Substation	Yes
J1566	Solar	Southern Minnesota Municipal Power Agency	NRIS	150	Rutland Substation 161kV	No
J1572	Hybrid	Northern States Power (Xcel Energy)	NRIS	150	Huntley - Blue Earth 161kV	Yes
J1575	Wind	Otter Tail Power Company	NRIS	70	Warsaw - Donaldson 115 kV	Yes
J1581	Solar	Northern States Power (Xcel Energy)	NRIS	200	Nobles County 345 kV Substation	Yes
J1582	Solar	Northern States Power (Xcel Energy)	NRIS	160	Pine Lake - Three Lakes 115 kV	Yes
J1588	Solar	Northern States Power (Xcel Energy)	NRIS	200	Bison 345kV	Yes
J1590	Wind	MidAmerican Energy Company	NRIS	200	O'Brien - Raun 345 kV	Yes
J1594	Wind	Otter Tail Power Company	NRIS	200	Big Stone - Ellendale 345 kV	Yes
J1597	Solar	Great River Energy	NRIS	75	Wing River Substation 230kV	No
J1605	Solar	Northern States Power (Xcel Energy)	NRIS	200	Sherburne 345kV	Yes
J1611	Solar	Minnesota Power (Allete Inc)	NRIS	174	Langola 115kV Tap	Yes
J1613	Wind	Northern States Power (Xcel Energy)	NRIS	400	Bison Substation 345 kV	Yes
J1620	Solar	Northern States Power (Xcel Energy)	NRIS	125	Pipestone to Split Rock 115kV	Yes
J1621	Wind	Otter Tail Power Company	NRIS	400	Big Stone South to Ellendale 345kV	Yes
J1622	Solar	Otter Tail Power Company	NRIS	100	Harvey-Underwood 230kV	Yes
J1657	Solar	Northern States Power (Xcel Energy)	NRIS	100	Pine Lake - Apple River 161 kV	Yes
J1730	Solar	Northern States Power (Xcel Energy)	NRIS	300	Eau Claire - AS King 345 kV	Yes
J1733	Battery Storage	ITC Midwest	NRIS	50	Lansing 161 kV	Yes
J1734	Solar	ITC Midwest	NRIS	150	Wever 161 kV	Yes
J1754	Wind	Great River Energy	NRIS	200	McHenry-Coal Creek Tap 230 kV	No

Project ID	Fuel Type	TO	Service Request Type	Service Request (MW)	POI	Interconnection Self Fund Election
J1766	Hybrid	Montana-Dakota Utilities Company	NRIS	20	Keystone XL 115kV	Yes
J1769	Wind	Dairyland Power Cooperative	NRIS	120	J898 POI 161 kV	Yes
J1801	Wind	ITC Midwest	NRIS Only	39	Story County Substation 161kV	Yes
J1804	Battery Storage	Northern States Power (Xcel Energy)	ERIS	0*	Pine Lake – Three Lakes 115 kV	Yes
J1808	Solar	Dairyland Power Cooperative	NRIS	200	Barron 161 kV	Yes
J1826	Hydro	Northern States Power (Xcel Energy)	NRIS	4	Cannon Falls 69 kV	Yes

*Denotes a Hybrid Application with another DPP 2020 Project



1.2. Project Summary Network Upgrades

J1494			
Network Upgrade	NU Cost Estimate (\$)	J1494 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$11,494	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$9,243	NRIS
Total Cost Per Project:		\$20,737	
J1495			
Network Upgrade	NU Cost Estimate (\$)	J1495 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$17,474	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$14,052	NRIS
Total Cost Per Project:		\$31,527	
J1496			
Network Upgrade	NU Cost Estimate (\$)	J1496 Cost Estimate (\$)	NU Type
Hazelton-Arnold 345 kV Structure Remediation	\$600,000	\$155,297	ERIS
Pleasant Valley 345/161 kV Transformer 1 Replacement	\$3,000,000	\$3,000,000	NRIS
Pleasant Valley 345/161 kV Transformer 2 Replacement	\$3,000,000	\$3,000,000	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$23,006	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$18,501	NRIS
Byron-Cascade 161 kV Line Reconductor	\$7,500,000	\$3,664,592	NRIS
Total Cost Per Project:		\$9,861,396	
J1498			
Network Upgrade	NU Cost Estimate (\$)	J1498 Cost Estimate (\$)	NU Type
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$16,031	ERIS
3rd Nobles 345/115 kV Transformer	\$10,000,000	\$4,483,759	NRIS
Pipestone-West Ridge 69 kV Rebuild	\$3,010,000	\$830,982	NRIS
West Ridge-Rock Tap 69 kV Rebuild	\$1,960,000	\$541,658	NRIS



J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$8,560	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$6,883	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$284,034	NRIS
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$870,091	OTP LPC
Total Cost Per Project:		\$7,041,998	
J1503			
Network Upgrade	NU Cost Estimate (\$)	J1503 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$25,660	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$20,635	NRIS
Total Cost Per Project:		\$46,295	
J1504			
Network Upgrade	NU Cost Estimate (\$)	J1504 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$25,660	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$20,635	NRIS
Total Cost Per Project:		\$46,295	
J1520			
Network Upgrade	NU Cost Estimate (\$)	J1520 Cost Estimate (\$)	NU Type
Glenworth-Hayward 161 kV Relay Upgrades	\$40,000	\$40,000	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$20,951	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$16,848	NRIS
Total Cost Per Project:		\$77,799	
J1521			
Network Upgrade	NU Cost Estimate (\$)	J1521 Cost Estimate (\$)	NU Type
Forman CT Replacement	\$50,000	\$17,216	ERIS
Hankinson-Wahpeton 230 kV Line Rebuild	\$22,000,000	\$11,917,526	ERIS
Hankinson-Forman 230 kV Line Rebuild	\$31,000,000	\$15,242,504	ERIS
Minn Valley-Granite Falls 230 kV Rebuild	\$4,500,000	\$850,187	ERIS
Wahpeton-Fergus Falls Structure Replacements	\$4,000,000	\$2,209,187	ERIS

Silver Lake-Fergus Falls Structure Replacements	\$1,050,000	\$713,737	ERIS
Silver Lake-Henning 230 kV Structure Remediation	\$500,000	\$339,875	ERIS
Aberdeen Jct-Ellendale 115 kV Terminal Upgrade	\$250,000	\$60,850	ERIS
GRE-Inman-GRE-Henning 230 kV Terminal Upgrade	\$40,000	\$27,190	ERIS
New J1594 POI-Big Stone South 345 kV Circuit	\$119,500,000	\$29,100,061	ERIS
Erie Junction 230 kV 4 steps of 50 MVar Capacitor Banks	\$6,000,000	\$823,529	ERIS
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$290,840	ERIS
Wing River 230 kV four steps of 50 MVar Capacitor Banks	\$3,750,000	\$523,973	ERIS
Hazel Creek 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$571,942	ERIS
MRES Alexandria six steps of 50 MVar Capacitor Banks	\$14,000,000	\$1,327,103	ERIS
Lyon County 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$581,717	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$283,212	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$6,602,960	NRIS
Hankinson 230 kV Terminal Upgrade	\$200,000	\$49,745	NRIS
Big Stone 230/115 Transformer 2 Replacement	\$3,500,000	\$887,904	NRIS
Highway 12-Ortonville 115 kV Structure Replacement	\$150,000	\$37,862	NRIS
Big Stone-Highway 12 115 kV Switch Replacement	\$250,000	\$63,103	NRIS
Wishek-Napoleon SW 230 kV Reconductor	\$7,550,000	\$1,836,486	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$36,350	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$29,232	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$647,937	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$547,011	NRIS
Browns Valley-New Effington 230 kV Rebuild	\$9,000,000	\$2,285,483	NRIS
Big Stone-Browns Valley 230 kV Rebuild	\$24,000,000	\$6,093,157	NRIS
Ellendale-Aberdeen Jct 115 kV Terminal Upgrades (MDU LPC)	\$450,000	\$113,356	MDU LPC
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$34,078,550	OTP LPC
Two Steps of 50 MVar Reactors at the J1594 POI Substation (OTP LPC)	\$4,000,000	\$998,124	OTP LPC
Hankinson-Forman 230 kV ERIS Upgrade Incremental (OTP LPC)	\$500,000	\$166,299	OTP LPC



Oakes-Forman 230 kV Rebuild (OTP LPC)	\$6,000,000	\$1,995,984	OTP LPC
Total Cost Per Project:		\$121,350,192	
J1528			
Network Upgrade	NU Cost Estimate (\$)	J1528 Cost Estimate (\$)	NU Type
Total Cost Per Project:		\$0	
J1529			
Network Upgrade	NU Cost Estimate (\$)	J1529 Cost Estimate (\$)	NU Type
J1529 POI - Auburn 69 kV Structure Replacment	\$200,000	\$200,000	NRIS
Auburn-SHRWOOD 69 kV Re-conductor	\$6,000,000	\$6,000,000	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$4253	NRIS
Total Cost Per Project:		\$6,204,253	
J1530			
Network Upgrade	NU Cost Estimate (\$)	J1530 Cost Estimate (\$)	NU Type
Creston-J1533 POI 161 kV Reconductor	\$10,500,000	\$202,440	ERIS
Macksburg-Winterset 161 kV Re-Conductor	\$10,000,000	\$289,372	NRIS
Total Cost Per Project:		\$491,812	
J1532			
Network Upgrade	NU Cost Estimate (\$)	J1532 Cost Estimate (\$)	NU Type
Creston-J1533 POI 161 kV Reconductor	\$10,500,000	\$483,073	ERIS
Winterset-Norwalk 161 kV Structure Replacements	\$400,000	\$27,826	NRIS
Macksburg-Winterset 161 kV Re-Conductor	\$10,000,000	\$693,847	NRIS
Total Cost Per Project:		\$1,204,746	
J1533			
Network Upgrade	NU Cost Estimate (\$)	J1533 Cost Estimate (\$)	NU Type
Creston-J1533 POI 161 kV Reconductor	\$10,500,000	\$9,814,487	ERIS
Adams-J1533 POI 161 kV Reconductor	\$2,500,000	\$2,500,000	ERIS

Clarinda-Brooks 161 kV Reconductor	\$13,000,000	\$13,000,000	ERIS
Brooks-Adams 161 kV Relay Settings	\$25,000	\$25,000	ERIS
Winterset-Norwalk 161 kV Structure Replacements	\$400,000	\$372,174	NRIS
Brooks-Clarinda 161 Terminal Upgrade	\$750,000	\$750,000	NRIS
Macksburg-Winterset 161 kV Re-Conductor	\$10,000,000	\$9,016,781	NRIS
Total Cost Per Project:		\$35,478,442	
J1534			
Network Upgrade	NU Cost Estimate (\$)	J1534 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$69,486	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$55,879	NRIS
Byron-Cascade 161 kV Line Reconductor	\$7,500,000	\$3,835,408	NRIS
Total Cost Per Project:		\$3,960,773	
J1544			
Network Upgrade	NU Cost Estimate (\$)	J1544 Cost Estimate (\$)	NU Type
Mandan-Mandan BN Line Rebuild	\$3,825,000	\$3,825,000	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$252,520	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$1,299,686	NRIS
Merricourt-Tatanka North 230 kV Re-Conductor	\$1,000,000	\$601,299	NRIS
Tatanka North-Foxtail 230 kV Substation Upgrade	\$1,500,000	\$902,832	NRIS
Foxtail-Ellendale 230 kV Re-Conductor	\$10,500,000	\$6,328,102	NRIS
Merricourt-Ellendale 230 kV Re-Conductor	\$18,000,000	\$10,848,274	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$34,837	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$28,014	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$495,245	NRIS
Sweet Ave-Midway 115 kV Rebuild (MDU LPC)	\$10,800,000	\$10,800,000	MDU LPC
26 th Exp-East Bismark 115 kV Terminal Upgrades (MDU LPC)	\$250,000	\$250,000	MDU LPC
East Bismark-Bismark 115 kV Terminal Upgrades (MDU LPC)	\$350,000	\$350,000	MDU LPC
Bisstein-NE Bismark 115 kV Terminal Upgrades (MDU LPC)	\$200,000	\$200,000	MDU



			LPC
Heskett-Mandan 115 kV Structure Remediation (MDU LPC)	\$200,000	\$200,000	MDU LPC
Mandan BN-Midway 115 kV Rebuild (MDU LPC)	\$900,000	\$900,000	MDU LPC
Total Cost Per Project:		\$37,315,809	
J1566			
Network Upgrade	NU Cost Estimate (\$)	J1566 Cost Estimate (\$)	NU Type
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$13,740	ERIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$26,950	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$21,672	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$357,499	NRIS
Murphy-Austin 161 kV Rebuild	\$2,000,000	\$2,000,000	NRIS
Total Cost Per Project:		\$2,419,861	
J1572			
Network Upgrade	NU Cost Estimate (\$)	J1572 Cost Estimate (\$)	NU Type
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$13,740	ERIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$28,856	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$23,205	NRIS
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$290,030	OTP LPC
Total Cost Per Project:		\$355,831	
J1575			
Network Upgrade	NU Cost Estimate (\$)	J1575 Cost Estimate (\$)	NU Type
J1575 POI-Donaldson 115 kV Rebuild	\$1,500,000	\$1,500,000	ERIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$329,315	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$13,923	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$11,197	NRIS
Donaldson-J1575 POI Terminal Equipment Upgrades (OTP LPC)	\$1,000,000	\$1,000,000	OTP LPC



Big Stone South-Alexandria 345 kV line	\$192,000,000	\$2,465,257	OTP LPC
Total Cost Per Project:		\$5,319,693	
J1581			
Network Upgrade	NU Cost Estimate (\$)	J1581 Cost Estimate (\$)	NU Type
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$27,481	ERIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$31,099	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$25,009	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$1,149,239	NRIS
Total Cost Per Project:		\$1,232,828	
J1582			
Network Upgrade	NU Cost Estimate (\$)	J1582 Cost Estimate (\$)	NU Type
Pine Lake Transformer Upgrade	\$2,500,000	\$2,025,475	NRIS
Three Lakes 115/69 kV Transformer Upgrade	\$2,500,000	\$2,500,000	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$35,210	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$28,315	NRIS
Total Cost Per Project:		\$4,589,000	
J1588			
Network Upgrade	NU Cost Estimate (\$)	J1588 Cost Estimate (\$)	NU Type
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$436,670	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$1,263,609	NRIS
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	\$848,160	NRIS
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	\$848,160	NRIS
Maple River -Sheyenne 230 kV Rebuild	\$16,000,000	\$3,200,000	NRIS
Inman-Wing River Jumper Replacement	\$40,000	\$8,558	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$39,920	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$32,102	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$465,443	NRIS
Total Cost Per Project:		\$7,142,622	
J1590			

Network Upgrade	NU Cost Estimate (\$)	J1590 Cost Estimate (\$)	NU Type
Raun-Remson 345 kV Structure Replacements	\$1,600,000	\$1,600,000	ERIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$4,523	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$3,637	NRIS
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$1,305,136	OTP LPC
Total Cost Per Project:		\$2,913,296	
J1594			
Network Upgrade	NU Cost Estimate (\$)	J1594 Cost Estimate (\$)	NU Type
Hankinson-Wahpeton 230 kV Line Rebuild	\$22,000,000	\$4,474,227	ERIS
Hankinson-Forman 230 kV Line Rebuild	\$31,000,000	\$7,865,140	ERIS
Minn Valley-Granite Falls 230 kV Rebuild	\$4,500,000	\$947,238	ERIS
Wahpeton-Fergus Falls Structure Replacements	\$4,000,000	\$770,318	ERIS
Aberdeen Jct-Ellendale 115 kV Terminal Upgrade	\$250,000	\$63,037	ERIS
New J1594 POI-Big Stone South 345 kV Circuit	\$119,500,000	\$30,117,121	ERIS
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$277,099	ERIS
Wing River 230 kV four steps of 50 MVar Capacitor Banks	\$3,750,000	\$174,658	ERIS
Hazel Creek 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$550,360	ERIS
MRES Alexandria six steps of 50 MVar Capacitor Banks	\$14,000,000	\$785,047	ERIS
Lyon County 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$581,717	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$283,461	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$7,082,133	NRIS
Hankinson 230 kV Terminal Upgrade	\$200,000	\$49,202	NRIS
Big Stone 230/115 Transformer 2 Replacement	\$3,500,000	\$949,972	NRIS
Highway 12-Ortonville 115 kV Structure Replacement	\$150,000	\$39,732	NRIS
Big Stone-Highway 12 115 kV Switch Replacement	\$250,000	\$66,220	NRIS
Wishek-Napoleon SW 230 kV Reconductor	\$7,550,000	\$1,641,868	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$36,481	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$29,337	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$656,309	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$581,170	NRIS

Browns Valley-New Effington 230 kV Rebuild	\$9,000,000	\$2,453,943	NRIS
Big Stone-Browns Valley 230 kV Rebuild	\$24,000,000	\$6,548,591	NRIS
Ellendale-Aberdeen Jct 115 kV Terminal Upgrades (MDU LPC)	\$450,000	\$109,323	MDU LPC
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$28,858,006	OTP LPC
Two Steps of 50 MVar Reactors at the J1594 POI Substation (OTP LPC)	\$4,000,000	\$1,000,625	OTP LPC
Total Cost Per Project:		\$96,992,335	
J1597			
Network Upgrade	NU Cost Estimate (\$)	J1597 Cost Estimate (\$)	NU Type
MN-STP-GRE-Aldrich-Verndale-Wing River 115 kV Line Remediation	\$6,300,000	\$6,300,000	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$164,438	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$492,123	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$15,792	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$11,768	NRIS
Wing River 230/115 kV Transformer Upgrade (GRE LPC)	\$3,000,000	\$3,000,000	GRE LPC
Total Cost Per Project:		\$9,984,121	
J1605			
Network Upgrade	NU Cost Estimate (\$)	J1605 Cost Estimate (\$)	NU Type
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$1,377,383	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$47,358	NRIS
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	\$38,084	NRIS
Total Cost Per Project:		\$1,462,825	
J1611			
Network Upgrade	NU Cost Estimate (\$)	J1611 Cost Estimate (\$)	NU Type
Little Falls - GRE-Dewing 115 kV Line Reconductor	\$14,000,000	\$14,000,000	ERIS
Fisher Hill-St. Stephen 115 kV Line Rebuild	\$6,900,000	\$6,900,000	ERIS



Fisher Hill-LeSauk Tap 115 kV Line Rebuild	\$3,000,000	\$3,000,000	ERIS
West St. Cloud-LeSauk Tap 115 kV Line Rebuild	\$1,300,000	\$1,300,000	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$760,056	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$39,172	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$31,501	NRIS
Total Cost Per Project:		\$26,030,730	
J1613			
Network Upgrade	NU Cost Estimate (\$)	J1613 Cost Estimate (\$)	NU Type
Minn Valley-Granite Falls 230 kV Rebuild	\$4,500,000	\$1,377,531	ERIS
Hubbard-Erie Jct 230 kV Structure Remediation	\$1,215,000	\$1,215,000	ERIS
Badura-Hubbard 230 kV Structure Remediation	\$2,700,000	\$2,700,000	ERIS
West St. Cloud 115/69 kV XFMR Replacment	\$1,939,875	\$1,939,875	ERIS
Sheyenne-Lake Park 230 Structure Remediation	\$350,000	\$350,000	ERIS
Erie Junction 230 kV 4 steps of 50 MVar Capacitor Banks	\$6,000,000	\$2,735,294	ERIS
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$283,969	ERIS
Wing River 230 kV four steps of 50 MVar Capacitor Banks	\$3,750,000	\$1,643,836	ERIS
Hazel Creek 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$544,964	ERIS
MRES Alexandria six steps of 50 MVar Capacitor Banks	\$14,000,000	\$7,457,944	ERIS
Lyon County 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$506,925	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$873,341	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$2,527,219	NRIS
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	\$3,392,639	NRIS
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	\$3,392,639	NRIS
Maple River -Sheyenne 230 kV Rebuild	\$16,000,000	\$12,800,000	NRIS
Inman-Wing River Jumper Replacement	\$40,000	\$17,117	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$79,840	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$64,205	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$930,887	NRIS
CSLTNET-Mapleton 115 kV Switch Replacement (OTP LPC)	\$200,000	\$200,000	OTP LPC
CSLTNET-Buffalo 115 kV Rebuild (OTP LPC)	\$4,500,000	\$4,500,000	OTP LPC

Buffalo 345/115 kV Transformer 1 Replacement	\$3,000,000	\$3,000,000	OTP LPC
Buffalo 345/115 kV Transformer 2 Replacement	\$3,000,000	\$3,000,000	OTP LPC
Mapleton-Sheyenne Rebuild (OTP LPC)	\$3,500,000	\$3,500,000	OTP LPC
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$13,921,450	OTP LPC
Erie Jct-Audubon 230 kV Rebuild (OTP LPC)	\$4,500,000	\$3,529,532	OTP LPC
Total Cost Per Project:		\$70,484,207	
J1620			
Network Upgrade	NU Cost Estimate (\$)	J1620 Cost Estimate (\$)	NU Type
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$27,481	ERIS
3rd Nobles 345/115 kV Transformer	\$10,000,000	\$5,516,241	NRIS
Pipestone-West Ridge 69 kV Rebuild	\$3,010,000	\$2,179,018	NRIS
West Ridge-Rock Tap 69 kV Rebuild	\$1,960,000	\$1,418,342	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$20,371	NRIS
Hazelon-Arnold 345 kV Sag Remediation	\$1,000,000	\$16,382	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$1,049,725	NRIS
Total Cost Per Project:		\$10,227,560	
J1621			
Network Upgrade	NU Cost Estimate (\$)	J1621 Cost Estimate (\$)	NU Type
Forman CT Replacement	\$50,000	\$32,784	ERIS
Hankinson-Wahpeton 230 kV Line Rebuild	\$22,000,000	\$5,608,247	ERIS
Hankinson-Forman 230 kV Line Rebuild	\$31,000,000	\$7,892,355	ERIS
Minn Valley-Granite Falls 230 kV Rebuild	\$4,500,000	\$666,483	ERIS
Forman 230/115 kV Transformer Replacement	\$3,000,000	\$3,000,000	ERIS
Second Ellendale 230/115 kV Transformer	\$5,009,000	\$5,009,000	ERIS
Wahpeton-Fergus Falls Structure Replacements	\$4,000,000	\$1,020,495	ERIS
Silver Lake-Fergus Falls Structure Replacements	\$1,050,000	\$336,263	ERIS

Silver Lake-Henning 230 kV Structure Remediation	\$500,000	\$160,125	ERIS
Aberdeen Jct-Ellendale 115 kV Terminal Upgrade	\$250,000	\$126,113	ERIS
GRE-Inman-GRE-Henning 230 kV Terminal Upgrade	\$40,000	\$12,810	ERIS
New J1594 POI-Big Stone South 345 kV Circuit	\$119,500,000	\$60,282,818	ERIS
Erie Junction 230 kV 4 steps of 50 MVar Capacitor Banks	\$6,000,000	\$1,558,824	ERIS
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$529,008	ERIS
Wing River 230 kV four steps of 50 MVar Capacitor Banks	\$3,750,000	\$1,160,959	ERIS
Hazel Creek 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$1,062,950	ERIS
MRES Alexandria six steps of 50 MVar Capacitor Banks	\$14,000,000	\$2,841,121	ERIS
Lyon County 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$1,072,022	ERIS
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	\$568,919	NRIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$12,384,482	NRIS
Hankinson 230 kV Terminal Upgrade	\$200,000	\$101,053	NRIS
Big Stone 230/115 Transformer 2 Replacement	\$3,500,000	\$1,662,124	NRIS
Highway 12-Ortonville 115 kV Structure Replacement	\$150,000	\$72,406	NRIS
Big Stone-Highway 12 115 kV Switch Replacement	\$250,000	\$120,677	NRIS
Inman-Wing River Jumper Replacement	\$40,000	\$14,325	NRIS
Wishek-Napoleon SW 230 kV Reconductor	\$7,550,000	\$4,071,645	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$72,813	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$58,554	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$1,285,829	NRIS
Split Rock 230 kV Transformer Upgrade	\$5,000,000	\$1,031,320	NRIS
Browns Valley-New Effington 230 kV Rebuild	\$9,000,000	\$4,260,575	NRIS
Big Stone-Browns Valley 230 kV Rebuild	\$24,000,000	\$11,358,252	NRIS
Ellendale-Aberdeen Jct 115 kV Terminal Upgrades (MDU LPC)	\$450,000	\$227,321	MDU LPC
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$99,335,347	OTP LPC
Two Steps of 50 MVar Reactors at the J1594 POI Substation (OTP LPC)	\$4,000,000	\$2,001,251	OTP LPC
Erie Jct-Audubon 230 kV Rebuild (OTP LPC)	\$4,500,000	\$970,468	OTP LPC
Hankinson-Forman 230 kV ERIS Upgrade Incremental (OTP LPC)	\$500,000	\$333,701	OTP

			LPC
Additional Lake Park-Sheyenne Structure Remediation (OTP LPC)	\$350,000	\$350,000	OTP LPC
Oakes-Forman 230 kV Rebuild (OTP LPC)	\$6,000,000	\$4,004,016	OTP LPC
Total Cost Per Project:		\$236,657,455	
J1622			
Network Upgrade	NU Cost Estimate (\$)	J1622 Cost Estimate (\$)	NU Type
Total Cost Per Project:		\$0	
J1657			
Network Upgrade	NU Cost Estimate (\$)	J1657 Cost Estimate (\$)	NU Type
Pine Lake Transformer Upgrade	\$2,500,000	\$474,525	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$21,380	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$17,193	NRIS
Total Cost Per Project:		\$513,098	
J1730			
Network Upgrade	NU Cost Estimate (\$)	J1730 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$68,533	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$55,112	NRIS
Total Cost Per Project:		\$123,645	
J1733			
Network Upgrade	NU Cost Estimate (\$)	J1733 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$4,878	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$3,923	NRIS
Seneca-Genoa 161 kV Rebuild	\$23,722,800	\$10,616,668	DPC LPC
Total Cost Per Project:		\$10,625,469	
J1734			



Network Upgrade	NU Cost Estimate (\$)	J1734 Cost Estimate (\$)	NU Type
Denmark-Newport 161 kV Structure Remediation	\$1,400,000	\$1,400,000	NRIS
Burlington-Denmark 161 kV Circuit 2 Rebuild	\$16,500,000	\$16,500,000	NRIS
Wever-Niota 161 kV Sag Remediation	\$100,000	\$100,000	NRIS
Wever-Burlington 161 kV Rebuild	\$9,600,000	\$9,600,000	NRIS
Total Cost Per Project:		\$27,600,000	
J1754			
Network Upgrade	NU Cost Estimate (\$)	J1754 Cost Estimate (\$)	NU Type
Minn Valley-Granite Falls 230 kV Rebuild	\$4,500,000	\$658,561	ERIS
Souris-Magic City 115 kV Rebuild	\$7,000,000	\$7,000,000	ERIS
McHenry 230/115 kV Transformer Replacement	\$6,057,149	\$6,057,149	ERIS
Drayton-Prairie Structure Replacement	\$50,000	\$50,000	ERIS
Erie Junction 230 kV 4 steps of 50 MVar Capacitor Banks	\$6,000,000	\$882,353	ERIS
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	\$20,611	ERIS
Wing River 230 kV four steps of 50 MVar Capacitor Banks	\$3,750,000	\$246,575	ERIS
Hazel Creek 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$269,784	ERIS
MRES Alexandria six steps of 50 MVar Capacitor Banks	\$14,000,000	\$1,588,785	ERIS
Lyon County 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	\$257,618	ERIS
Big Stone-Blair 230 kV Rebuild	\$33,000,000	\$1,018,473	NRIS
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	\$259,201	NRIS
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	\$259,201	NRIS
Merricourt-Tatanka North 230 kV Re-Conductor	\$1,000,000	\$365,014	NRIS
Tatanka North-Foxtail 230 kV Substation Upgrade	\$1,500,000	\$546,709	NRIS
Foxtail-Ellendale 230 kV Re-Conductor	\$10,500,000	\$3,819,343	NRIS
Merricourt-Ellendale 230 kV Re-Conductor	\$18,000,000	\$6,548,299	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$35,453	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$28,510	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$471,806	NRIS
Big Stone South-Alexandria 345 kV line	\$192,000,000	\$10,876,133	OTP LPC
Total Cost Per Project:		\$41,259,579	

J1766			
Network Upgrade	NU Cost Estimate (\$)	J1766 Cost Estimate (\$)	NU Type
Merricourt-Tatanka North 230 kV Re-Conductor	\$1,000,000	\$33,686	NRIS
Tatanka North-Foxtail 230 kV Substation Upgrade	\$1,500,000	\$50,460	NRIS
Foxtail-Ellendale 230 kV Re-Conductor	\$10,500,000	\$352,555	NRIS
Merricourt-Ellendale 230 kV Re-Conductor	\$18,000,000	\$603,427	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$3,177	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$2,555	NRIS
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	\$46,544	NRIS
Total Cost Per Project:		\$1,092,404	
J1769			
Network Upgrade	NU Cost Estimate (\$)	J1769 Cost Estimate (\$)	NU Type
Hazelton-Arnold 345 kV Structure Remediation	\$600,000	\$444,703	ERIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$39,770	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$31,982	NRIS
Byron-Pleasant Valley 161 kV Rebuild	\$24,900,000	\$24,900,000	NRIS
Seneca-Genoa 161 kV Rebuild	\$23,722,800	\$13,106,132	DPC LPC
Jericho-Alta Vis 69 kV Rebuild	\$521,310	\$521,310	DPC LPC
Jericho-Howard Tap 69 kV Rebuild	\$2,232,000	\$2,232,000	DPC LPC
Saratoga-Howard Tap 69 kV Rebuild	\$3,257,000	\$3,257,000	DPC LPC
Saratoga-Rice 69 kV Rebuild	\$263,000	\$263,000	DPC LPC
Total Cost Per Project:		\$44,795,897	
J1801			
Network Upgrade	NU Cost Estimate (\$)	J1801 Cost Estimate (\$)	NU Type
Traer-Abbott 161 kV Rebuild	\$19,060,000	\$19,060,000	NRIS

Total Cost Per Project:		\$19,060,000	
J1804			
Network Upgrade	NU Cost Estimate (\$)	J1804 Cost Estimate (\$)	NU Type
Total Cost Per Project:		0	
J1808			
Network Upgrade	NU Cost Estimate (\$)	J1808 Cost Estimate (\$)	NU Type
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$39,023	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$31,381	NRIS
Barron- Wash Co 161 kV Rebuild (DPC LPC)	\$29,044,000	\$29,044,000	DPC LPC
Elk Mound Wheaton 161 kV Jumper Replacement	\$50,000	\$50,000	DPC LPC
Wash Co-St Lake 161 kV Rebuild	\$26,298,600	\$26,298,600	DPC LPC
Total Cost Per Project:		\$55,463,004	
J1826			
Network Upgrade	NU Cost Estimate (\$)	J1808 Cost Estimate (\$)	NU Type
Cannon Falls-CNFLSTR 69 kV Rebuild	\$910,000	\$910,000	NRIS
Spring Creek 161/69 kV Transformer Upgrade	\$2,000,000	\$2,000,000	NRIS
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	\$691	NRIS
Hazelton-Arnold 345 kV Sag Remediation	\$1,000,000	\$556	NRIS
Total Cost Per Project:		\$2,911,248	

Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2020 West Phase 1 reflects responsibilities for mitigating system impacts. The total cost of network upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies.

Table 2: Total Cost of Network Upgrades for DPP 2020 West Phase 1 Projects



Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Steady State	LPC Studies	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
A	B	C	D	E	F	G	H	I	J	k = b+c+d+e+g+h+j	\$4,000/MW at Application	10% of (k) from Phase I-M2
J1494	\$0	\$0	Phase 2	Phase 2	Phase 2	\$20,737	\$3,500,000	\$0	\$0	\$3,520,737	\$200,000	\$152,074
J1495	\$0	\$0	Phase 2	Phase 2	Phase 2	\$31,527	\$4,000,000	\$0	\$0	\$4,031,527	\$200,000	\$203,153
J1496	\$155,297	\$0	Phase 2	Phase 2	Phase 2	\$9,706,099	\$3,000,000	\$0	\$0	\$12,861,396	\$200,000	\$1,086,140
J1498	\$16,031	\$870,091	Phase 2	Phase 2	Phase 2	\$6,155,876	\$3,500,000	\$0	\$0	\$10,541,998	\$200,000	\$854,200
J1503	\$0	\$0	Phase 2	Phase 2	Phase 2	\$46,295	\$4,500,000	\$0	\$0	\$4,546,295	\$600,000	\$0
J1504	\$0	\$0	Phase 2	Phase 2	Phase 2	\$46,295	\$4,500,000	\$0	\$0	\$4,546,295	\$600,000	\$0
J1520	\$0	\$0	Phase 2	Phase 2	Phase 2	\$77,798	\$10,000,000	\$0	\$0	\$10,077,798	\$540,000	\$467,780
J1521	\$64,597,438	\$37,352,313	Phase 2	Phase 2	Phase 2	\$19,400,443	\$4,000,000	\$0	\$0	\$125,350,194	\$798,000	\$11,737,019
J1528	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	\$0	\$0	\$0	\$0	\$0	\$0
J1529	\$0	\$0	Phase 2	Phase 2	Phase 2	\$6,204,253	\$12,000,000	\$0	\$0	\$18,204,253	\$200,000	\$1,620,425
J1530	\$202,440	\$0	Phase 2	Phase 2	Phase 2	\$289,372	\$10,000,000	\$0	\$0	\$10,491,812	\$200,000	\$849,181
J1532	\$483,073	\$0	Phase 2	Phase 2	Phase 2	\$721,673	\$12,000,000	\$0	\$0	\$13,204,746	\$200,000	\$1,120,475
J1533	\$25,339,487	\$0	Phase 2	Phase 2	Phase 2	\$10,138,955	\$13,000,000	\$0	\$0	\$48,478,442	\$800,000	\$4,047,844
J1534	\$0	\$0	Phase 2	Phase 2	Phase 2	\$3,960,773	\$0	\$0	\$0	\$3,960,773	\$400,000	\$0
J1544	\$3,825,000	\$12,700,000	Phase 2	Phase 2	Phase 2	\$20,790,810	\$6,000,000	\$0	\$0	\$43,315,810	\$800,000	\$3,531,581
J1566	\$13,740	\$0	Phase 2	Phase 2	Phase 2	\$2,406,121	\$2,500,000	\$0	\$0	\$4,919,861	\$600,000	\$0
J1572	\$13,740	\$290,030	Phase 2	Phase 2	Phase 2	\$52,061	\$10,000,000	\$0	\$0	\$10,355,831	\$600,000	\$435,583
J1575	\$1,500,000	\$3,465,257	Phase 2	Phase 2	Phase 2	\$354,436	\$4,000,000	\$0	\$0	\$9,319,693	\$280,000	\$651,969
J1581	\$27,481	\$0	Phase 2	Phase 2	Phase 2	\$1,205,346	\$6,500,000	\$0	\$0	\$7,732,827	\$800,000	\$0
J1582	\$0	\$0	Phase 2	Phase 2	Phase 2	\$4,589,000	\$9,000,000	\$0	\$0	\$13,589,000	\$640,000	\$718,900
J1588	\$0	\$0	Phase 2	Phase 2	Phase 2	\$7,142,623	\$6,500,000	\$0	\$0	\$13,642,623	\$800,000	\$564,262
J1590	\$1,600,000	\$1,305,136	Phase 2	Phase 2	Phase 2	\$8,160	\$2,500,000	\$0	\$0	\$5,413,296	\$800,000	\$0
J1594	\$46,605,962	\$29,967,954	Phase 2	Phase 2	Phase 2	\$20,418,419	\$4,000,000	\$0	\$0	\$100,992,335	\$800,000	\$9,299,234
J1597	\$6,300,000	\$3,000,000	Phase 2	Phase 2	Phase 2	\$684,121	\$1,600,000	\$0	\$0	\$11,584,121	\$400,000	\$758,412
J1605	\$0	\$0	Phase 2	Phase 2	Phase 2	\$1,462,825	\$6,500,000	\$0	\$0	\$7,962,825	\$800,000	\$0
J1611	\$25,200,000	\$0	Phase 2	Phase 2	Phase 2	\$830,730	\$8,300,000	\$0	\$0	\$34,330,730	\$696,000	\$2,737,073

Project	ERIS Network Upgrades (\$)					NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Steady State	LPC Studies	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
A	B	C	D	E	F	G	H	I	J	k = b+c+d+e+g+h+j	\$4,000/MW at Application	10% of (k) from Phase I-M2
J1613	\$20,755,338	\$31,650,982	Phase 2	Phase 2	Phase 2	\$24,077,886	\$6,500,000	\$0	\$0	\$82,984,206	\$1,600,000	\$6,698,421
J1620	\$27,481	\$0	Phase 2	Phase 2	Phase 2	\$10,200,080	\$3,500,000	\$0	\$0	\$13,727,561	\$500,000	\$872,756
J1621	\$92,372,377	\$107,222,104	Phase 2	Phase 2	Phase 2	\$37,062,974	\$8,000,000	\$0	\$0	\$244,657,455	\$1,600,000	\$22,865,746
J1622	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	\$0	\$0	\$0	\$0	\$400,000	\$0
J1657	\$0	\$0	Phase 2	Phase 2	Phase 2	\$513,099	\$10,000,000	\$0	\$0	\$10,513,099	\$400,000	\$651,310
J1730	\$0	\$0	Phase 2	Phase 2	Phase 2	\$123,645	\$13,000,000	\$0	\$0	\$13,123,645	\$1,200,000	\$112,365
J1733	\$0	\$10,616,668	Phase 2	Phase 2	Phase 2	\$8,800	\$1,250,000	\$0	\$0	\$11,875,468	\$200,000	\$987,547
J1734	\$0	\$0	Phase 2	Phase 2	Phase 2	\$27,600,000	\$1,250,000	\$0	\$0	\$28,850,000	\$600,000	\$2,285,000
J1754	\$17,031,436	\$10,876,133	Phase 2	Phase 2	Phase 2	\$13,352,010	\$7,000,000	\$0	\$0	\$48,259,579	\$800,000	\$4,025,958
J1766	\$0	\$0	Phase 2	Phase 2	Phase 2	\$1,092,404	\$6,000,000	\$0	\$0	\$7,092,404	\$80,000	\$629,240
J1769	\$444,703	\$19,379,442	Phase 2	Phase 2	Phase 2	\$24,971,753	\$750,000	\$0	\$0	\$45,595,898	\$480,000	\$4,079,590
J1801	\$0	N/A	Phase 2	Phase 2	Phase 2	\$19,060,000	N/A	\$0	\$0	\$19,060,000	\$156,000	\$1,750,000
J1804	\$0	\$0	Phase 2	Phase 2	Phase 2	\$0	N/A	\$0	\$0	\$0	\$0	\$0
J1808	\$0	\$55,392,600	Phase 2	Phase 2	Phase 2	\$70,404	\$4,300,000	\$0	\$0	\$59,763,004	\$800,000	\$5,176,300
J1826	\$0	\$0	Phase 2	Phase 2	Phase 2	\$2,911,248	\$2,000,000	\$0	\$0	\$4,911,248	\$16,000	\$475,125
Total	\$306,511,024	\$324,088,710	Phase 2	Phase 2	Phase 2	\$277,789,052	\$265,450,000	\$0	\$0	\$1,088,450,076	\$21,886,000	\$86,152,219

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. Energy Resource Interconnection Service (ERIS) Network Upgrades and Network Resource Interconnection Service (NRIS) Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.

Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
Second Ellendale 230/115 kV Transformer	\$5,009,000	MDU	ERIS	Yes
Wahpeton-Fergus Falls Structure Replacements	\$4,000,000	OTP	ERIS	Yes
Silver Lake-Fergus Falls Structure Replacements	\$1,050,000	OTP/MRES	ERIS	Yes

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
Raun-Remson 345 kV Structure Replacements	\$1,600,000	MEC	ERIS	Yes
Silver Lake-Henning 230 kV Structure Remediation	\$500,000	OTP/MRES	ERIS	Yes
Sheyenne-Lake Park 230 Structure Remediation	\$350,000	OTP	ERIS	Yes
Drayton-Prairie Structure Replacement	\$50,000	OTP	ERIS	Yes
Aberdeen Jct-Ellendale 115 kV Terminal Upgrade	\$250,000	MDU	ERIS	Yes
GRE-Inman-GRE-Henning 230 kV Terminal Upgrade	\$40,000	GRE	ERIS	No
Creston-J1533 POI 161 kV Reconductor	\$10,500,000	MEC	ERIS	Yes
Adams-J1533 POI 161 kV Reconductor	\$2,500,000	MEC	ERIS	Yes
Clarinda-Brooks 161 kV Reconductor	\$13,000,000	MEC	ERIS	Yes
Brooks-Adams 161 kV Relay Settings	\$25,000	MEC	ERIS	Yes
MN-STP-GRE-Aldrich-Verndale-Wing River 115 kV Line Remediation	\$6,300,000	MP/GRE	ERIS	MP Yes GRE No
Mandan-Mandan BN Line Rebuild	\$3,825,000	MDU	ERIS	Yes
Little Falls - GRE-Dewing 115 kV Line Reconductor	\$14,000,000	MP/GRE	ERIS	Yes
Fisher Hill-St. Stephen 115 kV Line Rebuild	\$6,900,000	MP	ERIS	Yes
Fisher Hill-LeSauk Tap 115 kV Line Rebuild	\$3,000,000	MP	ERIS	Yes
West St. Cloud-LeSauk Tap 115 kV Line Rebuild	\$1,300,000	XEL	ERIS	Yes
New J1594 POI-Big Stone South 345 kV Circuit	\$119,500,000	MDU/OTP	ERIS	Yes
J1575 POI-Donaldson 115 kV Rebuild	\$1,500,000	OTP	ERIS	Yes
Erie Junction 230 kV 4 steps of 50 MVar Capacitor Banks	\$6,000,000	OTP	ERIS	Yes
Maynard 115 kV two steps of 40 MVar Capacitor Banks	\$1,500,000	XEL	ERIS	Yes
Wing River 230 kV four steps of 50 MVar Capacitor Banks	\$3,750,000	GRE	ERIS	No
Hazel Creek 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	XEL	ERIS	Yes
MRES Alexandria six steps of 50 MVar Capacitor Banks	\$14,000,000	MRES	ERIS	Yes
Lyon County 345 kV two steps of 40 MVar Capacitor Banks	\$3,000,000	XEL	ERIS	Yes
Pine Lake Transformer Upgrade	\$2,500,000	XEL	NRIS	Yes
3rd Nobles 345/115 kV Transformer	\$10,000,000	XEL	NRIS	Yes
Coon Creek-Kohlman Lake 345 kV Terminal Upgrade	\$5,000,000	XEL	NRIS	Yes
Cannon Falls-CNFLSTR 69 kV Rebuild	\$910,000	XEL	NRIS	Yes

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
Pipestone-West Ridge 69 kV Rebuild	\$3,010,000	XEL	NRIS	Yes
West Ridge-Rock Tap 69 kV Rebuild	\$1,960,000	XEL	NRIS	Yes
Pleasant Valley 345/161 kV Transformer 1 Replacement	\$3,000,000	GRE	NRIS	No
Pleasant Valley 345/161 kV Transformer 2 Replacement	\$3,000,000	GRE	NRIS	No
J1529 POI - Auburn 69 kV Structure Replacement	\$200,000	MEC	NRIS	Yes
Three Lakes 115/69 kV Transformer Upgrade	\$2,500,000	XEL	NRIS	Yes
Spring Creek 161/69 kV Transformer Upgrade	\$2,000,000	GRE	NRIS	No
Big Stone-Blair 230 kV Rebuild	\$33,000,000	OTP/NWE	NRIS	Yes
Hankinson 230 kV Terminal Upgrade	\$200,000	OTP	NRIS	Yes
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	OTP	NRIS	Yes
Maple River 345/230 kV Transformer 1 Replacement	\$4,500,000	OTP	NRIS	Yes
Winterset-Norwalk 161 kV Structure Replacements	\$400,000	MEC	NRIS	Yes
Denmark-Newport 161 kV Structure Remediation	\$1,400,000	MEC	NRIS	Yes
Burlington-Denmark 161 kV Circuti 2 Rebuild	\$16,500,000	MEC	NRIS	Yes
Glenworth-Hayward 161 kV Relay Upgrades	\$40,000	ITCM	NRIS	Yes
Wever-Niota 161 kV Sag Remediation	\$100,000	MEC	NRIS	Yes
Brooks-Clarinda 161 Terminal Upgrade	\$750,000	MEC	NRIS	Yes
Auburn-SHRWOOD 69 kV Re-conductor	\$6,000,000	MEC	NRIS	Yes
Macksburg-Winterset 161 kV Re-Conductor	\$10,000,000	MEC	NRIS	Yes
Maple River -Sheyenne 230 kV Rebuild	\$16,000,000	XEL	NRIS	Yes
Big Stone 230/115 Transformer 2 Replacement	\$3,500,000	OTP	NRIS	Yes
Highway 12-Ortonville 115 kV Structure Replacement	\$150,000	OTP	NRIS	Yes
Big Stone-Highway 12 115 kV Switch Replacement	\$250,000	OTP	NRIS	Yes
Inman-Wing River Jumper Replacement	\$40,000	GRE	NRIS	No
Wishek-Napoleon SW 230 kV Reconductor	\$7,550,000	MDU	NRIS	Yes
Merricourt-Tatanka North 230 kV Re-Conductor	\$1,000,000	MDU	NRIS	Yes
Tatanka North-Foxtail 230 kV Substation Upgrade	\$1,500,000	MDU	NRIS	Yes
Foxtail-Ellendale 230 kV Re-Conductor	\$10,500,000	MDU	NRIS	Yes
Merricourt-Ellendale 230 kV Re-Conductor	\$18,000,000	MDU	NRIS	Yes
J1181 POI-J1413 POI 345 Sag Remediation	\$1,000,000	ITCM	NRIS	Yes

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
Hazleton-Arnold 345 kV Sag Remediation	\$1,000,000	ITCM	NRIS	Yes
Traer-Abbott 161 kV Rebuild	\$19,060,000	ITCM	NRIS	Yes
Wever-Burlington 161 kV Rebuild	\$9,600,000	ITCM	NRIS	Yes
Granite Falls-MN Valley Tap 230 kV Rebuild	\$5,000,000	XEL	NRIS	Yes
Split Rock 230 kV Transformer Upgrade	\$5,000,000	XEL	NRIS	Yes
Browns Valley-New Effington 230 kV Rebuild	\$9,000,000	MDU	NRIS	Yes
Byron-Cascade 161 kV Line Reconductor	\$7,500,000	SMMPA	NRIS	No
Murphy-Austin 161 kV Rebuild	\$2,000,000	SMMPA	NRIS	No
Big Stone-Browns Valley 230 kV Rebuild	\$24,000,000	MDU	NRIS	Yes
Sweet Ave-Midway 115 kV Rebuild (MDU LPC)	\$10,800,000	MDU	LPC	Yes
26 th Exp-East Bismark 115 kV Terminal Upgrades (MDU LPC)	\$250,000	MDU	LPC	Yes
Ellendale-Aberdeen Jct 115 kV Terminal Upgrades (MDU LPC)	\$450,000	MDU	LPC	Yes
East Bismark-Bismark 115 kV Terminal Upgrades (MDU LPC)	\$350,000	MDU	LPC	Yes
Bisstein-NE Bismark 115 kV Terminal Upgrades (MDU LPC)	\$200,000	MDU	LPC	Yes
Heskett-Mandan 115 kV Structure Remediation (MDU LPC)	\$200,000	MDU	LPC	Yes
Mandan BN-Midway 115 kV Rebuild (MDU LPC)	\$900,000	MDU	LPC	Yes
Two Steps of 50 MVAR Reactors at the J1594 POI Substation (OTP LPC)	\$4,000,000	OTP	LPC	Yes
CSLTNET-Mapleton 115 kV Switch Replacement (OTP LPC)	\$200,000	OTP	LPC	Yes
CSLTNET-Buffalo 115 kV Rebuild (OTP LPC)	\$4,500,000	OTP	LPC	Yes
Buffalo 345/115 kV Transformer 1 Replacement	\$3,000,000	OTP	LPC	Yes
Buffalo 345/115 kV Transformer 2 Replacement	\$3,000,000	OTP	LPC	Yes
Mapleton-Sheyenne Rebuild (OTP LPC)	\$3,500,000	OTP	LPC	Yes
Donaldson-J1575 POI Terminal Equipment Upgrades (OTP LPC)	\$1,000,000	OTP	LPC	Yes
Erie Jct-Audubon 230 kV Rebuild (OTP LPC)	\$4,500,000	OTP	LPC	Yes
Hankinson-Forman 230 kV ERIS Upgrade Incremental (OTP LPC)	\$500,000	OTP	LPC	Yes
Lake Park-Sheyenne 230 kV Structure Replacements (OTP LPC)	\$350,000	OTP	LPC	Yes
Oakes-Forman 230 kV Rebuild (OTP LPC)	\$6,000,000	OTP	LPC	Yes
Big Stone South-Alexandria 345 kV line	\$192,000,000	OTP/MRES	LPC	Yes
Wing River 230/115 kV Transformer Upgrade (GRE LPC)	\$3,000,000	GRE	LPC	No

Network Upgrade	Cost (\$)	TO	Upgrade Type	Self Fund
Barron- Wash Co 161 kV Rebuild (DPC LPC)	\$29,044,000	DPC	LPC	Yes
Elk Mound Wheaton 161 kV Jumper Replacement	\$50,000	DPC	LPC	Yes
Jericho-Alta Vis 69 kV Rebuild	\$521,310	DPC	LPC	Yes
Jericho-Howard Tap 69 kV Rebuild	\$2,232,000	DPC	LPC	Yes
Saratoga-Howard Tap 69 kV Rebuild	\$3,257,000	DPC	LPC	Yes
Saratoga-Rice 69 kV Rebuild	\$263,000	DPC	LPC	Yes
Seneca-Genoa 161 kV Rebuild	\$23,722,800	DPC	LPC	Yes
Wash Co-St Lake 161 kV Rebuild	\$26,298,600	DPC	LPC	Yes

Table 4: Shared Network Upgrades (Planning level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
None				

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, stated that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all non-synchronous projects included in the DPP 2020 West study cycle.

In this study, the power factor at the high-side of the generator substation for each project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that not all projects meet FERC Order 827 requirements. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1494	52.8	+/- 38.184	N/A	50	28.3	0.870	49.7	54.8	0.672	Yes	N/A
J1495	52.8	+/- 38.184	N/A	50	28.3	0.870	49.7	54.8	0.672	Yes	N/A
J1496	52.8	+/- 38.184	N/A	50	28.3	0.870	49.7	54.8	0.672	Yes	N/A
J1498	52.8	+/-20.9	Single Step 7 MVar	50	22.3	0.913	50	27.8	0.874	Yes	N/A
J1503	150	+/-72.9	Single Step 4 MVar	146.1	50	0.946	143.7	121.5	0.764	Yes	N/A
J1504	150	+/-72.9	Single Step 4 MVar	146.1	50	0.946	143.7	121.5	0.764	Yes	N/A
J1520	135	+/-65.4	Single Step 23 MVar	131.8	41.1	0.955	129.9	110.8	0.761	No	2
J1521	199.5	+/-96.2	Single Step 4 MVar	195.4	87.8	0.912	191.2	180.9	0.726	Yes	N/A

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensati on	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1528*	76.6	+/-47.3	Four Steps 6 MVar	75.5	70.4	0.731	75.2	52.9	0.818	Yes	N/A
J1529	50.12	+/-24.22	Two Steps 5 MVar	49.7	19.6	0.930	49.5	31.3	0.845	Yes	N/A
J1530	50.12	+/-24.22	Two Steps 5 MVar	49.7	19.6	0.930	49.5	31.3	0.845	Yes	N/A
J1532	50.12	+/-24.22	Two Steps 5 MVar	49.7	19.6	0.930	49.5	31.3	0.845	Yes	N/A
J1533	205	+/- 154.224	N/A	199. 8	127	0.844	196	229. 8	0.649	Yes	N/A
J1534	102	+/-78.2	N/A	100. 3	58	0.866	99	114	0.656	Yes	N/A
J1544	200	+/-65	N/A	196. 1	16.1	0.997	192.8	156. 3	0.777	No	48
J1566	182.4	+/-50	Eight Steps 5 MVar	150. 4	78.6	0.886	150	78.6	0.886	Yes	N/A
J1572	227.15	+/- 139.818	N/A	149. 8	104.3	0.821	146.9	230. 4	0.538	Yes	N/A
J1575	72.8	+34.6/- 25.94	Two Steps 5 MVar	72.2	32.8	0.910	71.9	41	0.869	Yes	N/A
J1581	243.2	+/-66	Six Steps 7 MVar	200. 4	67	0.948	198.6	141. 6	0.814	Yes	N/A
J1582**	165.2	+/-82.38	Two Steps 6 MVar	160	59.7	0.937	157.6	94.6	0.857	Yes	N/A
J1588	243.2	+/-66	Eight Steps 4 MVar	200. 2	60.4	0.957	198.6	122. 6	0.851	No	5
J1590	200	+/-96.9	Two Steps 6 MVar	196. 9	73.3	0.937	196.3	137. 2	0.820	Yes	N/A
J1594	206.72	+/- 100.12	N/A	200. 7	65.8	0.950	198.9	161. 6	0.776	Yes	N/A

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1597	75.9	+/-16.5	Four Steps 3.5 MVar	74.9	14.2	0.982	74.5	40.2	0.880	No	10
J1605	243.2	+/-66	Six Steps 7 MVar	200.4	67	0.948	198.6	141.6	0.814	Yes	N/A
J1611	212.8	+/-58	Eight Steps 7 MVar	199.8	60.6	0.957	198	136	0.824	No	5
J1613	413.44	+/-200.24	N/A	408	130	0.953	400	217	0.879	No	4
J1620	125	+/-41.08	Two Steps 16 MVar	123.8	64.6	0.887	123.2	66	0.881	Yes	N/A
J1621	413.44	+/-200.24	N/A	408	130	0.953	400	217	0.879	No	4
J1622	102	+/-78.19	N/A	100.3	58	0.866	99	114	0.656	Yes	N/A
J1657	100.75	+/-52.482	N/A	98.9	41.7	0.921	97.8	75.8	0.790	Yes	N/A
J1730*	310.1	+/-181.8	Four Steps 6 MVar	297	97.9	0.950	285.8	263	0.736	Yes	N/A
J1733	50	+/-24.7	N/A	49.6	16	0.952	49.4	37.5	0.797	No	1
J1734	150	+/-72.8	N/A	146.1	50	0.946	143.7	121.5	0.764	Yes	N/A
J1754	208	+42.236 /-60.667	N/A	202.3	7.2	0.999	197.9	121.8	0.852	No	59
J1766	21	+/-6.55	One Step of 1.7 MVar	20.1	6.7	0.949	20	8.7	0.917	Yes	N/A
J1769	120	+/-49.908	One Step 8 MVar	118.3	35.4	0.958	117.4	85	0.810	No	3
J1801***	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
J1804**	40.8	+/-25.3	Two Steps 6 MVar	40.4	40.1	0.710	40.4	25.5	0.846	Yes	N/A
J1808	200	+/-101	N/A	194.9	54.7	0.963	192.1	122.4	0.843	No	9
J1826	4.4	+/-4.85	N/A	4	4.3	0.681	3.9	5.7	0.565	Yes	N/A

*Hybrid Interconnection Requests

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensati on	VAR Injection			VAR Absorption			FERC Order 827 Compliant	MVar Deficiency
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVA r)	Leading p.f. (pu)		
**Hybrid Interconnection Requests											
***NRIS Only Request											

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2020 West models is based on the MTEP 20 series models with the Bench Cases including all prior-queued projects and their associated network upgrades.

- Bench Cases
 - DPP20-2025SH90-PhaseI-Bench_Final
 - DPP20-2025SUM-PhaseI-Bench_Final
- Study Cases
 - DPP20-2025SUM-PhaseI-Study_Final
 - DPP20-2025SH90-PhaseI-Study_Final
 - DPP20-2025SH90-PhaseI-Study_Charging_Final

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2020 West generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1801d.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or

- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner FERC filed Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the Deliverability algorithm, to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Backbone Network Upgrade Analysis

Significant thermal overloads were observed for the outages related to Big Stone-Twin Brooks as well as Twin Brooks-J1594 POI. This issue has been observed in previous studies in which generators propose tapping the Big Stone-Ellendale 345 kV line. A new line from J1594-Big Stone 345 kV was needed to address these observed issues. Appendix C contain detail results from the analysis.

5. Thermal Analysis

The thermal analysis results for 2020 West group show generator projects causing constraints. The details pertaining to the thermal analysis can be found in Appendix A — .

6. Voltage Analysis

The voltage analysis results for summer peak models show that voltage violations are observed in both the peak and off peak scenarios. Results of this analysis are included in Appendix B. A voltage stability analysis was also performed on the MWEX interface. A detailed report is in Appendix E.

7. Stability Analysis

A voltage stability analysis to be performed in Phase 2.

8. Short Circuit Analysis

Short circuit analysis to be performed in Phase 2.

9. Affected System Impact Study

Affected System analysis to be performed in Phase 2.

10. Deliverability Analysis

10.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. Any

generator that has a 5% DFAX on any constrained flowgate is undeliverable. If the generator is deemed not fully deliverable, the customer, at decision point, can choose either to change the project to an Energy Resource (ER) project or nominate a lower NRIS amount. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting “bottled up”.

MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

10.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

10.3. Deliverability Study Results

The deliverability analysis assumes all ERIIS upgrades as those upgrades are required for the study group to interconnection to the transmission system. With all the MISO analysis **ERIS upgrades assumed as well as the prior queued network upgrades**, the deliverability analysis determined NRIS service amounts that can be granted with no network upgrades. If a constraint is identified, the deliverability is calculated by taking the NRIS Request amount subtracting the calculated shared deductible. The shared deduction is the amount by which each study generator affecting a given flowgate must be scaled down in order to prevent loading violations on the flowgate. The deduction is proportionally taken from each study generator affecting a particular flowgate. If a particular monitored element becomes a flowgate under multiple contingencies, the contingency with the highest total shared deduction is used. Detailed tables on the breakdown of NRIS amounts with network upgrades are in Appendix F.

J1494 is deliverable to 28.75 MWs without network upgrades
J1495 is deliverable to 28.75 MWs without network upgrades
J1496 is deliverable to 0 MWs without network upgrades
J1498 is deliverable to 0 MWs without network upgrades
J1503 is deliverable to 86.24 MWs without network upgrades
J1504 is deliverable to 86.24 MWs without network upgrades
J1520 is deliverable to 0 MWs without network upgrades
J1521 is deliverable to 0 MWs without network upgrades
J1528 did not request NRIS
J1529 is deliverable to 0 MWs without network upgrades
J1530 is deliverable to 21.61 MWs without network upgrades
J1532 is deliverable to 34.62 MWs without network upgrades
J1533 is deliverable to 22.92 MWs without network upgrades
J1534 is deliverable to 91.68 MWs without network upgrades
J1544 is deliverable to 0 MWs without network upgrades
J1566 is deliverable to 60.04 MWs without network upgrades
J1572 is deliverable to 86.24 MWs without network upgrades
J1575 is deliverable to 37.93 MWs without network upgrades
J1581 is deliverable to 101.62 MWs without network upgrades
J1582 is deliverable to 22.86 MWs without network upgrades
J1588 is deliverable to 128.68 MWs without network upgrades
J1590 is deliverable to 23 MWs without network upgrades
J1594 is deliverable to 0 MWs without network upgrades
J1597 is deliverable to 54.2 MWs without network upgrades

J1605 is deliverable to 115 MWs without network upgrades
 J1611 is deliverable to 100.4 MWs without network upgrades
 J1613 is deliverable to 216.7 MWs without network upgrades
 J1620 is deliverable to 0 MWs without network upgrades
 J1621 is deliverable to 0 MWs without network upgrades
 J1622 is deliverable to 100 MWs without network upgrades
 J1657 is deliverable to 57.5 MWs without network upgrades
 J1730 is deliverable to 172.49 MWs without network upgrades
 J1733 is deliverable to 29.89 MWs without network upgrades
 J1734 is deliverable to 0 MWs without network upgrades
 J1754 is deliverable to 0 MWs without network upgrades
 J1766 is deliverable to 0 MWs without network upgrades
 J1769 is deliverable to 0 MWs without network upgrades
 J1801 is deliverable to 0 MWs without network upgrades
 J1804 did not request NRIS
 J1808 is deliverable to 114.99 MWs without network upgrades
 J1826 is deliverable to 0 MWs without network upgrades

11. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis tests for Network Upgrades driven by higher queued interconnection projects was performed for this System Impact Study. SNUs were not identified for DPP 2020 West Area requests.

12. Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Interconnection service as of the draft System Impact Study report date.

12.1. Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade identified in System Impact Study was provided by the corresponding transmission owning company.

12.2. Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all Study Group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any network upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A cost portion of NU}$$

$$\text{Cost of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.

Cost Allocation Methodology for Voltage Constraints

Cost allocation of voltage constraint driven network upgrades will be determined by the pro rata share of the voltage impact each project has on the most constrained bus under the most constraining contingency. The voltage impact of each project will be calculated by locking all voltage regulating equipment in the model and then backing out each GI project one at a time to identify each project's impact to the constraint. In severe instances of voltage collapse where projects cannot be backed out one at a time, they will be added one at a time to determine their impact to the constraint.

As the number and types of constraints increases, mitigating the constraints individually may result in higher overall costs. In instances when mitigation(s) resolve multiple types of constraints (such as thermal + voltage or thermal + voltage + transient stability) the cost is allocated based off the ratio share of the total cost of the independent mitigations in order to equitably allocate the cost to all parties contributing to constraints. In summary, only the lowest cost mitigation option will be constructed, but for cost allocation purposes the independent mitigations are required.

Appendix A – Thermal Analysis Results

Thermal Constraints

Appendix B – Voltage Constraints

Voltage Constraints

Appendix C – Backbone Analysis Results

Appendix D – Network Upgrade Cost Allocation

Appendix E – MWEX Voltage Stability Report

Appendix F – Deliverability Tables

Appendix G – Deliverability Results

Appendix H – Transmission Owner LPC Studies